

Taller 1, punto 5: Dataset_train_test modificado para banknote

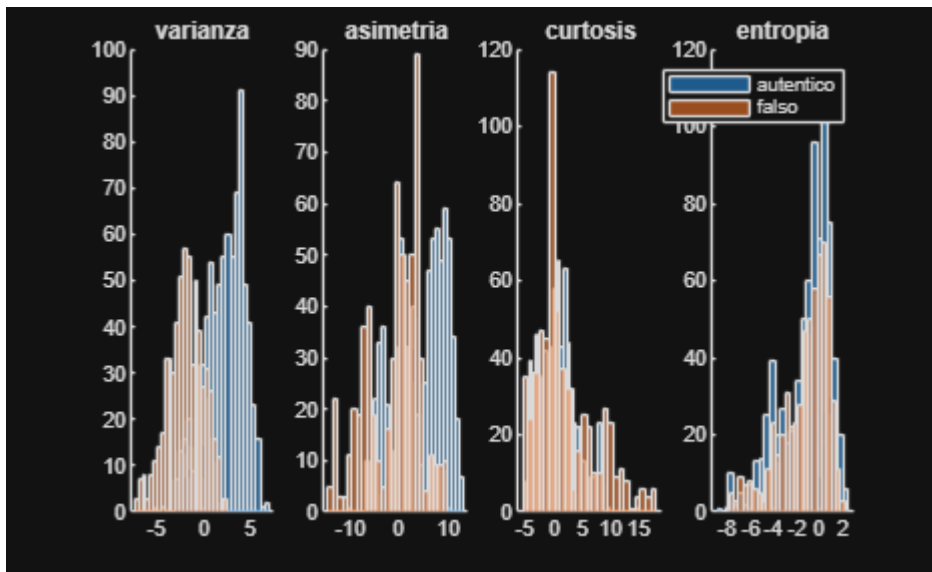
1372 billetes, 4 atributos de la transformada wavelet de la imagen, clase 0 autentico 1 falso

```
clear; close all; clc

datos = load('data_banknote_authentication.txt');
No_examples = size(datos, 1);
```

Un vistazo a las variables

```
nombres = {'varianza', 'asimetria', 'curtosis', 'entropia'};
figure
for j = 1:4
    subplot(1,4,j); hold on
    histogram(datos(datos(:,5)==0, j), 25)
    histogram(datos(datos(:,5)==1, j), 25)
    title(nombres{j})
end
legend('autentico', 'falso')
```



Se mezcla con semilla fija

```
rng(2021)
ind = randperm(No_examples);
```

Division 60-40, 70-30, 80-20 y 90-10

aqui si hace falta el round: $0.9 \times 1372 = 1234.8$

```
proporciones = [0.6 0.7 0.8 0.9];
for k = 1:4
```

```

n_train = round(proporciones(k) * No_examples);
S(k).p   = proporciones(k);
S(k).Xtr = datos(ind(1:n_train), 1:4);
S(k).ytr = datos(ind(1:n_train), 5);
S(k).Xte = datos(ind(n_train+1:end), 1:4);
S(k).yte = datos(ind(n_train+1:end), 5);

```

```
end
```

```
S
```

```
S = 1x4 struct
```

Fields	p	Xtr	ytr	Xte	yte
1	0.6000	823×4 double	823×1 double	549×4 double	549×1 double
2	0.7000	960×4 double	960×1 double	412×4 double	412×1 double
3	0.8000	1098×4 double	1098×1 double	274×4 double	274×1 double
4	0.9000	1235×4 double	1235×1 double	137×4 double	137×1 double

```
save('banknote_splits.mat', 'S')
```